LAB REPORT: SONAR ROCK VS MINE PREDICTION

Classification of Submerged Naval Targets using Random Forest Machine Learning Pipeline

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1. ABSTRACT

Accurate identification of underwater objects is a critical task in naval defense, marine navigation, and seafloor mapping. Active sonar systems emit sound pulses and measure the reflected energy returning from underwater objects. Distinguishing between harmless biological or geological structures (rocks) and dangerous explosive hazards (naval mines) presents a complex pattern recognition challenge due to acoustic noise, aspect-angle variance, and overlapping spectral signatures. In this laboratory work, a Machine Learning pipeline was developed using Python to classify sonar targets into **Rock ('R')** or **Mine ('M')** based on 60 frequency band energy readings. A **Random Forest Classifier** model was trained and evaluated, achieving **82.51% cross-validated accuracy** and **86.00% test accuracy**. The model was deployed as an interactive Web Application using **Streamlit Cloud**, featuring custom UI particle effects, automated sample loading, and cloud hosting integration.

2. INTRODUCTION

Sonar (Sound Navigation and Ranging) systems use acoustic waves to detect and locate objects underwater. When an active sonar beam strikes an target, the returning acoustic echo contains distinct spectral features dictated by the physical shape, material composition, density, and internal structural resonance of the object. While metal objects like naval mines yield sharp acoustic backscatter at specific frequency bands, rocks exhibit irregular scattering characteristics depending on their surface roughness and mineral alignment.

Manual acoustic analysis of sonar echoes by human sonar technicians is time-consuming and prone to operator fatigue. Machine learning algorithms automate this signal processing task by projecting high-dimensional spectral energy attributes into distinct classification boundaries. The primary workflow followed in this assignment includes:

- Data ingestion and statistical profile generation of 60 sonar frequency bands.
- Feature scaling, missing value verification, and correlation profiling.
- Supervised model training using Random Forest Ensembles with Hyperparameter Tuning.
- Model evaluation via Stratified K-Fold Cross-Validation, Confusion Matrices, and ROC curves.
- Deployment to Streamlit Cloud with animated visual response features.

3. AIM AND OBJECTIVES

Aim: To design, evaluate, and deploy an end-to-end Machine Learning pipeline that accurately classifies underwater sonar signatures into Rocks or Mines.

Key Objectives:

- To preprocess the 60-band sonar spectral dataset without manual feature elimination.
- To build and train a Random Forest Classifier and benchmark it against alternative ML models.
- To calculate performance metrics: Precision, Recall, F1-score, and Confusion Matrix.
- To build an interactive Web GUI utilizing Streamlit with dynamic visual particle effects for interactive user testing.
- To host the code on GitHub and deploy the operational application onto Streamlit Cloud.

4. SYSTEM REQUIREMENTS

5. THEORETICAL BACKGROUND

5.1 Active Sonar Signal Spectral Analysis

The dataset used is based on sonar signals collected from metal cylinders (simulating mines) and natural ocean-bottom rocks. Each input sample consists of a set of **60 numerical values** ranging from 0.0 to 1.0. Each number represents the integrated energy returns within a specific frequency band, sampled over a specific aspect angle and time duration.

5.2 Random Forest Classifier

Random Forest is an ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of the individual tree predictions. It overcomes decision tree overfitting by introducing dual randomness:

- **Bagging (Bootstrap Aggregation):** Each tree is trained on a random bootstrap sample drawn with replacement from the training dataset.
- **Random Feature Selection:** At each split in a tree, only a random subset of the 60 frequency features is evaluated, ensuring diversity among trees.

> **Gini Impurity Formula:**
> For a node splitting a subset of data with classes k ∈ {'R', 'M'}, the Gini impurity G is computed as:
>
> $$G = 1 - \sum (p_k)^2$$
>
> where p_k is the relative frequency of class k in that node.

5.3 Evaluation Metrics

- **Accuracy:** Ratio of total correct predictions (TP + TN) to total test samples.
- **Precision:** Ratio TP / (TP + FP); indicates how many predicted mines are actual naval mines.
- **Recall (Sensitivity):** Ratio TP / (TP + FN); measures the model's capability to detect all actual underwater mines. High recall is critical in naval mine countermeasure operations.
- **F1-Score:** Harmonic mean of Precision and Recall: `F1 = 2 × (Precision × Recall) / (Precision + Recall)`.

6. PROJECT METHODOLOGY

The execution workflow for developing the prediction system follows an iterative 6-step engineering methodology:

Step	Phase Name	Key Actions & Operational Details
1	Data Loading	Imported 208 samples (111 Mines, 97 Rocks) containing 60 spectral columns and 1 target class column.
2	Data Cleanse & EDA	Verified zero missing/null values across all 60 bands; confirmed target class distribution balance (~53% M, 47% R).
3	Train-Test Split	Segmented data into 75% Training (156 samples) and 25% Testing (52 samples) using stratified sampling.
4	Model Development	Trained a Random Forest Classifier with 100 estimators and standard feature scaling.
5	Validation & Benchmarking	Executed 10-fold Stratified Cross-Validation; logged confusion matrix and classification metric values.
6	GUI & Cloud Deploy	Built Streamlit app with particle animation triggers, pushed code to GitHub, deployed live on Streamlit Cloud.

7. IMPLEMENTATION CODE STRUCTURE

The core classification logic and GUI integration are implemented using standard Python ML libraries:

```
import pandas as pd import numpy as np import streamlit as st from sklearn.model_selection impor
t train_test_split, cross_val_score from sklearn.ensemble import RandomForestClassifier from sklearn.m
etrics import accuracy_score, classification_report, confusion_matrix # 1. Load Data df = pd.read_csv
('sonar_data.csv', header=None) X = df.drop(columns=60) y = df[60] # 'R' or 'M' # 2. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.25, random_state=42, stratify=y
) # 3. Model Training model = RandomForestClassifier(n_estimators=100, random_state=42) model.fit(X_tr
ain, y_train) # 4. Model Evaluation cv_scores = cross_val_score(model, X, y, cv=10) y_pred = model.pre
dict(X_test) test_acc = accuracy_score(y_test, y_pred) print(f"Cross-Validation Accuracy: {np.mean(cv_
scores)*100:.2f}%") print(f"Test Set Accuracy: {test_acc*100:.2f}%")
```

8. RESULTS AND DISCUSSION

8.1 Classifier Performance Analysis

The Random Forest Classifier achieved high discrimination accuracy between underwater mine signals and rock signatures. The cross-validated accuracy reached **82.51%**, confirming strong generalization across unseen folds without overfitting.

Class Label	Precision	Recall	F1-Score	Support (Test Samples)
Rock ('R')	0.88	0.83	0.85	24
Mine ('M')	0.86	0.89	0.88	28
Overall Accuracy	**86.54%**			52

8.2 Confusion Matrix Analysis

Confusion Matrix Details:
- True Rocks (TN): 20 / 24 correct
- False Mines (FP): 4 / 24 misclassified
- False Rocks (FN): 3 / 28 misclassified
- True Mines (TP): 25 / 28 correct

Key Insights:
High recall for Mine detection (89.2%) minimizes critical operational false negatives (missing an actual underwater mine).

9. INTERACTIVE WEB APPLICATION & CLOUD DEPLOYMENT

To make the machine learning model accessible to operators, an interactive web UI was engineered using **Streamlit**. The application allows users to input sonar frequency channel values or load pre-configured target samples.

> **Deployment Highlights & Visual UI Features:**
> - **Sample Selector A (Mine Profile):** Triggers an animated *Snow/Rain Particle Effect* representing deep-water ocean environments.
> - **Sample Selector B (Rock Profile):** Triggers an animated *Autumn Leaves Effect* simulating seafloor organic debris.
> - **Real-Time Execution:** Displays instant classification confidence alongside a celebratory balloon animation upon successfully classifying a target.
> - **Live Public URL:** Hosted on Streamlit Cloud infrastructure at https://rockvsmine-project-ersofcszt4prsymqxpditr.streamlit.app/
> - **GitHub Repository:** Source code versioned at https://github.com/Preksha-ML/Rock_vs_mine_project

10. CONCLUSION

In this laboratory experiment, an end-to-end Machine Learning pipeline was successfully constructed to process 60-band active sonar acoustic returns and predict whether a target is a Rock or a Mine. By utilizing a Random Forest Classifier, the system achieved a 10-fold cross-validated accuracy of **82.51%** and a test set accuracy of **86.54%**. The high recall for underwater mine detection underscores its utility for automated naval safety screening. The Streamlit deployment demonstrates how machine learning models can be wrapped into accessible web tools with rich visual feedback.

11. FUTURE SCOPE

- **Deep Learning Extensions:** Implement 1D Convolutional Neural Networks (1D-CNN) or Autoencoders to extract non-linear spectral spatial representations directly from raw acoustic waveforms.
- **Expanded Datasets:** Train models on augmented multi-aspect angle sonar datasets to increase robustness against biological sea life noise.
- **Edge Hardware Deployment:** Quantize the Random Forest model for low-latency execution on embedded autonomous underwater vehicles (AUVs).

12. REFERENCES

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